Transferring Learned Repairs Between Fine-Tunes

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I compare two ways to reuse a repair learned for an earlier fine-tune: fit scalar coefficients over its fixed directions, or use it to initialize a repair whose directions can change. On two fine-tunes of a 67M-parameter language model, trained on different samples from the same data distribution, I repair the Dutch damage caused by training the model to predict German less accurately. At 2,048 fit tokens and a 600-step cap, donor-initialized repairs recover 11.3 percentage points (pp) more damage than the tested scalar-only search, 95% CI [+10.0, +12.7], on the primary target and 20.4 pp [+19.0, +21.7] on the replication target. Both exceed the prespecified 3 pp margin. Donor initialization is ahead of fresh fitting by 1.1 and 1.4 pp at the same budget, inside the prespecified ±2 pp equivalence band and comparable to the difference between two fresh seeds. Its advantage is larger with less data and fewer steps. On the tested grid, it reaches the quality of the first fresh seed with fewer fit-token presentations, while both families reach the reference with the same minimum number of unique tokens. For this donor, model and fine-tuning recipe, the learned repair helps most as an initialization when fitting data and steps are limited.

1 Introduction

Fine-tuning can damage behaviour outside the intended task. Repeating the same recipe on new samples can reproduce that damage, so a repair learned for an earlier fine-tune may help repair the next one. The earlier fine-tune is the *donor* and its repair is the *donor repair*. We can keep the donor repair's directions fixed and fit new scalar coefficients, or use it to initialize a repair whose directions can change.

The fine-tuning procedure deliberately worsens German next-token prediction and, as a side effect, raises Dutch cross-entropy (CE), a measure of prediction loss. This provides a controlled test: the repair must restore Dutch prediction quality while preserving the intended German change. I train the donor and two target fine-tunes independently from the same base model, using the same recipe on different samples from the same Europarl data distribution. The donor repair has one rank-one update per weight matrix. The primary comparison asks how much Dutch damage is recovered by rescaling its fixed directions and by allowing those directions to change. The descriptive secondary comparisons ask how donor initialization performs against fresh and random starts at each fit budget and step cap, and what it saves (Table 1).

I fit rank-one repairs with a KL objective on the model used in Goodfire's adVersarial Parameter Decomposition (VPD) study (Bushnaq et al. 2026). The repairs operate directly on its weight matrices, independently of the decomposition.

Several methods reuse a weight update with fixed or learned scales. Task arithmetic adds or subtracts a weight difference with a single scale (Ilharco et al. 2023). RESTA adds one separately obtained safety vector to fine-tunes in several domains, using a fixed global scale in its main experiments (Bhardwaj et al. 2024). SafeGene keeps a distilled safety adapter frozen and learns one scalar per selected layer, improving on a coefficient of 1 (Wang et al. 2026, sec. 3.5 and 4.4). Such reuse can also fail. EnchTable's preprint Table II reports a medical fine-tune that RESTA leaves less safe than before (Wu et al. 2026). Other work uses a learned object to initialize new training. In SPoT, soft prompts learned on source tasks initialize the prompt of a target task (Vu et al. 2022). These studies do not compare the two uses of one transferred update on the same target at matched rank and cost. This study makes that comparison at matched rank and reports the costs of each family.

2 Setup

Model and fine-tunes. The base model is Goodfire's four-layer decoder-only model of 67M parameters, trained on a subset of the Pile (Bushnaq et al. 2026, sec. 3.1; Gao et al. 2020). Each fine-tune trains an additive update D to the 24 weight matrices of the attention and MLP blocks. Embeddings and norms stay frozen. A hinge term raises German CE, and a KL term keeps English, French, Spanish and Italian, the *retain* languages, close to the base model. The text is Europarl session text (Koehn 2005). The donor fine-tune and the two targets follow the same recipe with different seeds and training dates (method details). Dutch is held out of fine-tuning. On the readout dates the primary target's Dutch damage is 0.187 nat/token [0.178, 0.196] and the replication target's 0.189 [0.180, 0.198].

Repair. A repair adds $E_m = b_m a_m^\top$ to each of the 24 matrices of the fine-tuned model, so it has rank one per matrix. The vector a_m is the input factor and b_m is the output factor. The repair minimizes KL divergence from two teacher models across three groups of text. The fine-tuned model is the teacher on German, to preserve the intended effect, and on the retain languages. The base model is the teacher on Dutch, to undo the damage. Each term uses $D_{\mathrm{KL}}(p_{\text{teacher}} \,\|\, p_{\text{repair}})$, averaged over the group's predicted positions. The loss sums the mean KL of the three groups with equal weights. The repair norm is capped at the fine-tune norm, $\|E\|_F \le \|D\|_F$. Every family trains full-batch for 600 steps at four learning rates and keeps checkpoints as candidates for selection. Families initialized from the donor repair scale it down if needed to meet the norm cap (method details).

Recovery. Recovery is the share of the fine-tune's Dutch damage that a repair X removes,

$$r_{\mathrm{nl}}(X) = 1 - \frac{\mathrm{CE}_{\mathrm{nl}}(X) - \mathrm{CE}_{\mathrm{nl}}(\text{base})}{\mathrm{CE}_{\mathrm{nl}}(\text{fine-tuned}) - \mathrm{CE}_{\mathrm{nl}}(\text{base})}.$$

Differences in recovery are reported in percentage points (pp).

In the scalar-only family, each donor repair update stays fixed and the repair is $E_m = c_m b_m a_m^\top$. Fitting the coefficients c_m trains 24 parameters. The other families train one or both factors, as listed in Table 1.

Table 1: Repair families. All have rank one per matrix and share the objective, norm cap, fit tokens, steps and selection rule. The number of trained parameters varies by family despite their equal rank. The fixed donor repair is included as a reference: it is capped and held fixed.

Family	Start	Trained (parameters)	What the comparison tests
Scalar coefficients only	capped donor repair	scalar coefficients c_m (24)	whether the donor repair's fixed directions suffice after rescaling
Donor initialization	capped donor repair	both factors (55,296)	whether training the factors improves on scalar fitting, and whether the donor repair is a useful start
Donor initialization, output only	capped donor repair	output vectors b_m (27,648)	how much of the gap closes while keeping the donor repair's input vectors fixed
Random initialization	random directions, same norm as the donor start	both factors (55,296)	whether the donor repair's directions help beyond the norm of its start
Fresh	$b_m = 0$, $a_m \sim \mathcal{N}(0, 1/d_{\text{in}})$	both factors (55,296)	whether the donor repair is a better start than fitting a new repair

Varying the fit budget and the step cap separates a gain in final recovery from a gain in reaching it sooner. The output-only and fresh families have counterparts in low-rank adapters. The output factor of an adapter matters more than its input factor when the input factor is random and frozen (Zhu et al. 2024). Here, the output-only family's frozen input vectors are the donor repair's trained ones. Fresh fitting uses the zero-output initialization of such adapters (Hu et al. 2022). A second seed of fresh and of random initialization, at the full budget only, shows how far a result moves between initializations. The first fresh seed starts from the same input vectors used to initialize the donor repair, while the second uses an independent draw.

Budgets and step caps. Each target has nested fit budgets of 256, 512 and 2,048 unique predicted tokens with a 2:1:1 split of Dutch, German and retain text. These budgets count prediction positions, excluding the initial

context token of each window. A separate selection and diagnostic pool contains 57,344 prediction positions, shared by all families. Checkpoints at steps ≤50, ≤100 and ≤200 of the same runs form step-capped cells, alongside the full 600-step runs. Each family therefore has a grid of 3 budgets × 4 step caps. The cost appendix gives the costs of selection and of the additional scalar optimization.

Selection and readout. For every family and cell, a fixed rule chooses one candidate on separate selection dates: among the candidates that keep the fine-tune's German effect and move Dutch toward the base model, it prefers higher Dutch recovery, subject to a guard against raising the CE of the retain languages (method details). The chosen repairs are read out once on 92 session dates from the validation period, separate from the project's earlier data windows (data details). The selected checkpoint can precede the cell's step cap. All cells of a target share one bootstrap over readout dates (Efron 1979), so contrasts are paired. Intervals are 95% percentile intervals.

Additional scalar optimization. At the 2,048-token budget, two additional runs optimize the scalar coefficients to convergence on the exact full-batch fit loss from two starts. The selection rule then chooses among the 48 training checkpoints and these 2 converged candidates. Both kinds are eligible, including the ordinary training checkpoints (appendix).

Interpretation rules. The primary contrast compares donor initialization with the scalar-only search with full-batch optimization at 2,048 tokens and 600 steps. An advantage of more than 3 pp is established if its whole interval lies above 3 pp, excluded if the whole interval lies below 3 pp, and undecided otherwise. I interpret this contrast on a target only if its Dutch damage is clear (the lower end of the readout interval is at least 0.025 nat/token), both selected repairs satisfy the readout constraints in Table 6, and scalar optimization converged from both starts. All other contrasts are descriptive, without multiplicity correction. Their intervals determine whether the sign is positive, negative or undecided. Five contrasts, marked in Table 3, are also assessed for equivalence within a ±2 pp band. The protocol fixes both margins without a stated derivation.

3 Results

3.1 Fixed directions fall short

Both targets meet the conditions for interpreting the primary contrast (Table 6 and Table 7). Donor initialization recovers 11.3 pp [+10.0, +12.7] more damage than the scalar-only search with full-batch optimization on the primary target (Table 2) and 20.4 pp [+19.0, +21.7] more on the replication target. Both intervals lie entirely above the 3 pp margin.

The gap is nearly as large at 256 and 512 tokens, where scalar fitting received no additional optimization to convergence (Figure 1). Scalar-only fitting recovers about 87% of the damage on the primary target and 77% on the replication target, and adding the converged candidates changes this little (Table 3). Every family that trains the factors recovers at least 95.6%. The shortfall is already present on the scalar-only repair's own fit tokens (appendix). Training only the output vectors, with the donor repair's input vectors fixed, closes most of the gap (Table 3). Training both factors recovers only slightly more.

Table 2: Primary target, 2,048 fit tokens, 92 readout dates. 95% intervals from a joint bootstrap over dates (5,000 draws). Donor-initialized fitting exceeds the tested scalar-coefficient search by more than 3 percentage points. The scalar search includes 4 learning rates × 600 steps plus two full-batch optimizations, with 50 candidates in total. Selection considers both ordinary training checkpoints and converged candidates.

Repair	Recovery of Dutch damage (%)	95% CI
Donor initialization (both factors trained)	98.2	[97.6, 98.9]
Scalar coefficients only + full-batch optimization	86.9	[85.5, 88.3]
Difference (percentage points)	+11.3	[+10.0, +12.7]

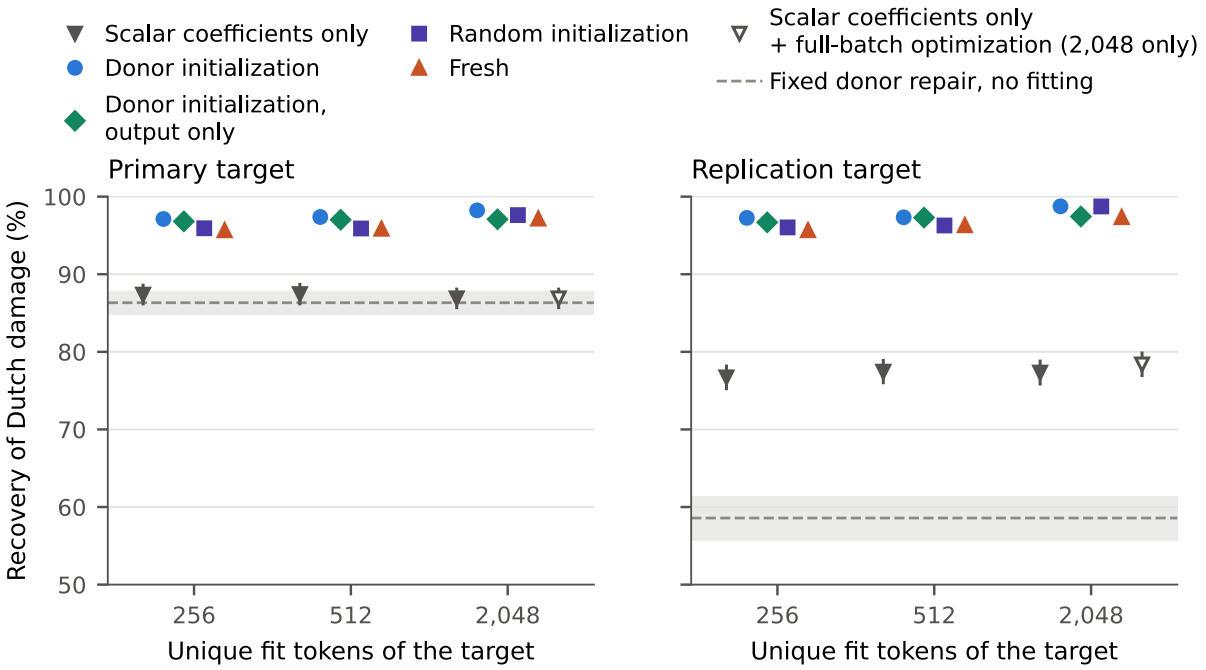


Figure 1: Recovery of the Dutch damage on the readout dates, selected from runs of up to 600 steps for each family and fit budget, with 95% bootstrap intervals over 92 dates (most are smaller than the markers). The band is the fixed donor repair and its interval. Scalar-only fitting falls well short of every family that trains the factors, at every budget and on both targets. Including the converged full-batch candidates in selection at 2,048 tokens (hollow marker) does not close the gap.

3.2 The donor as an initialization

At the full budget, donor initialization is ahead of fresh fitting by +1.1 pp [+0.8, +1.3] on the primary target and +1.4 pp [+1.1, +1.7] on the replication target (Table 3). Both intervals lie above zero and inside the ±2 pp band, so the advantage is positive but small enough to count as practical equivalence. It is comparable to the difference between two fresh seeds (+1.0 and +0.7 pp). Against the second fresh seed, donor initialization differs by only +0.04 pp [−0.19, +0.26] on the primary target and +0.7 pp [+0.5, +1.0] on the replication target.

Compared with random initialization of the same norm, donor initialization is ahead at the full budget on the primary target. The sign is undecided on the replication target (Table 3). The donor repair's directions help more in short runs with little data (Figure 2). With 256 fit tokens and at most 50 steps, donor initialization beats random initialization by +4.9 and +4.5 pp, and fresh fitting by +4.4 and +4.1 pp. These advantages are larger than at the full budget. Only one fresh seed and one random seed were tested at this small budget.

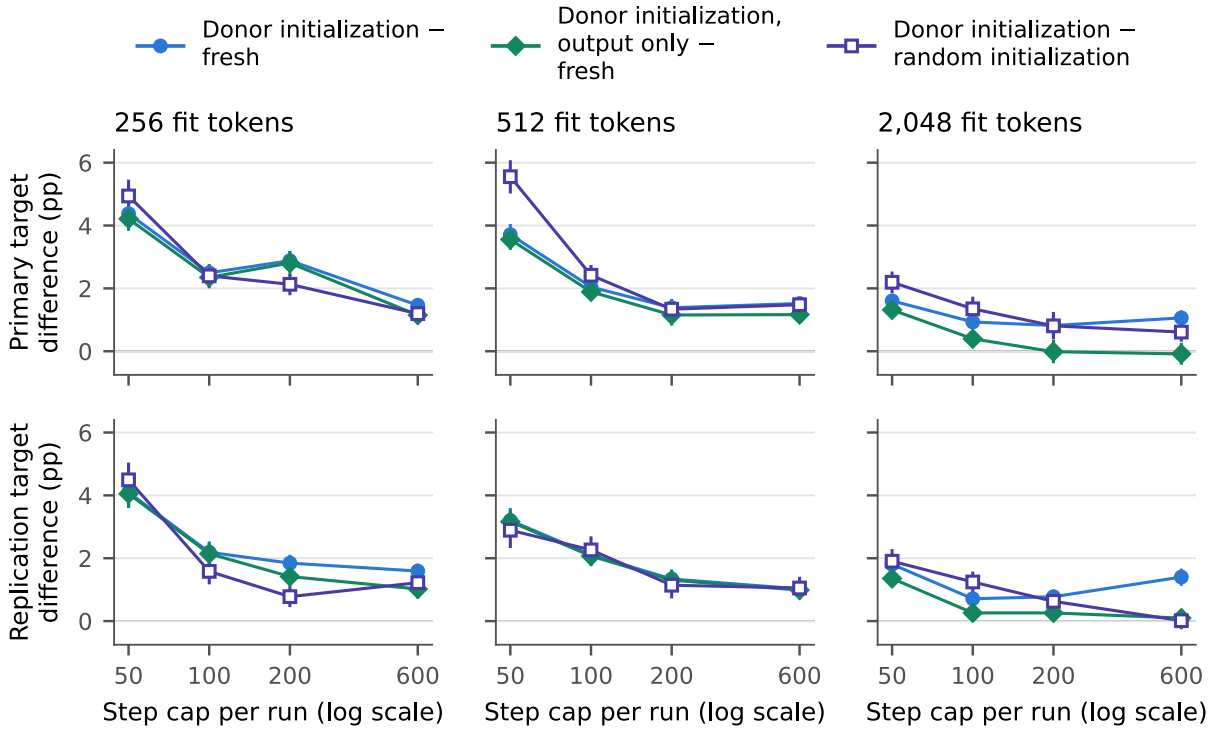


Figure 2: Paired differences of recovery at matched fit budgets and step caps: both families of a contrast have the same fit budget (columns) and step cap (horizontal axis, logarithmic, points at their true step caps), on the primary (top) and replication (bottom) target, with 95% bootstrap intervals over 92 dates for each fixed target. Donor initialization helps most in short runs on little data. With a 600-step cap and 2,048 tokens its advantage over fresh is about 1 pp and over random initialization between 0 and 1 pp.

The full-budget contrasts in Table 3 include output-only fitting and the second fresh and random seeds.

Table 3: Secondary contrasts at the full budget (2,048 unique fit tokens, up to 600 steps): paired difference in recovery (pp) with its 95% interval, 92 readout dates per target. Rows marked '+ full-batch optimization' also include the two converged candidates. These intervals are descriptive, without multiplicity correction. The last column applies only to the five contrasts named in the protocol for an equivalence reading (primary / replication): 'yes' means that the whole interval lies inside the ±2 pp band.

Contrast (2,048 fit tokens)	Primary target	Replication target	Inside ±2 pp
Donor initialization, output only − scalar coefficients only + full-batch optimization	+10.2 [+8.9, +11.5]	+19.1 [+17.6, +20.5]	—
Donor initialization − donor initialization, output only	+1.1 [+0.7, +1.5]	+1.3 [+1.0, +1.6]	yes / yes
Donor initialization − fresh	+1.1 [+0.8, +1.3]	+1.4 [+1.1, +1.7]	yes / yes
Donor initialization, output only − fresh	−0.1 [−0.4, +0.3]	+0.1 [−0.2, +0.4]	—
Donor initialization − random initialization	+0.6 [+0.3, +0.9]	+0.01 [−0.26, +0.28]	—
Random initialization − fresh	+0.5 [+0.2, +0.7]	+1.4 [+1.0, +1.7]	yes / yes
Scalar coefficients only + full-batch optimization − fresh	−10.3 [−11.7, −8.9]	−19.0 [−20.4, −17.5]	—
Scalar coefficients only + full-batch optimization − fixed donor repair	+0.6 [+0.1, +1.1]	+19.8 [+17.8, +22.0]	—
Scalar coefficients only + full-batch optimization − scalar coefficients only	same repair selected	+1.1 [+0.8, +1.3]	—

Contrast (2,048 fit tokens)	Primary target	Replication target	Inside ±2 pp
Fresh, second seed − fresh	+1.0 [+0.7, +1.3]	+0.7 [+0.4, +0.9]	yes / yes
Random initialization, second seed − random initialization	−0.38 [−0.76, −0.02]	−1.2 [−1.5, −0.9]	yes / yes
Donor initialization − fresh, second seed	+0.04 [−0.19, +0.26]	+0.7 [+0.5, +1.0]	—

3.3 What donor initialization saves

To compare fitting costs at similar recovery, I use the first fresh seed at 2,048 tokens with a 600-step cap as a reference. A grid cell *reaches* it if the selected repair satisfies the constraints and the lower end of its paired interval against the reference lies above −2 pp. Fit-token presentations count the optimizer steps of all four learning-rate runs multiplied by the unique fit tokens.

On the tested grid, donor initialization reaches this reference with one eighth of the fit-token presentations. Counted in presentations, the one-time donor cost is amortized after 11 targets. This is a descriptive projection from two tested targets; the intervals cover individual cells, not the selected cost ratio.

The cheapest reaching cells by unique-token cost use the same number of tokens: fresh fitting also reaches the reference, narrowly, with 256 fit tokens under a 600-step cap. In GPU seconds, break-even comes much later (about 151 and 106 targets). These seconds estimate allocated shared work from different runs; a saving claim would require more than these estimates.

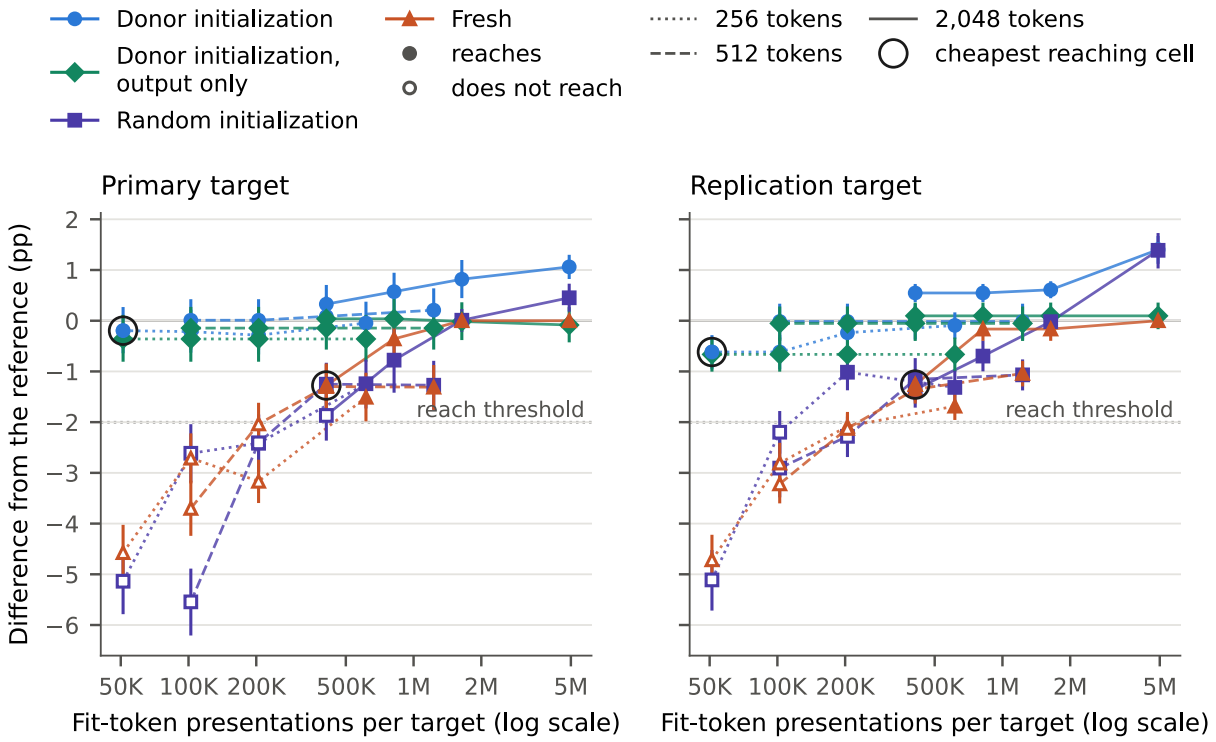


Figure 3: Quality against cost on the primary (left) and replication (right) target. Each point is a grid cell at its fit-token presentations, with its paired difference from the reference, fresh at 2,048 tokens with a 600-step cap, and its 95% bootstrap interval over 92 dates for each fixed target. Lines join the step caps 50, 100, 200 and 600. Filled markers reach the reference: chosen inside the constraints, interval above −2 pp (dotted line). Rings mark the cheapest reaching cells used for the break-even estimate: donor initialization at 256 tokens and fresh at 2,048 tokens, both with a 50-step cap (51,200 and 409,600 presentations). Scalar-only cells never reach and are not shown.

Table 4: Target-side cost of the cheapest cell of fresh fitting and of donor initialization that reaches the reference (fresh at 2,048 tokens and up to 600 steps), found separately in each currency. A cell is written as unique fit tokens / step cap. K* = donor cost / (fresh cost − donor-initialization cost) is a modelled amortization: reuse pays off for more than K* targets. The donor's one-time cost and what each currency counts and leaves out are in the cost-accounting appendix. GPU seconds are an estimate of allocated shared work, and the donor's come from a different run.

Currency	Target	Fresh	Fresh cell	Donor init.	Donor init. cell	Break-even K*
Unique tokens	Primary	57,600	256 / 600	57,600	256 / 600	no per-target saving
	Replication	57,600	256 / 600	57,600	256 / 600	no per-target saving
Fit-token presentations	Primary	409,600	2,048 / ≤50	51,200	256 / ≤50	10.3
	Replication	409,600	2,048 / ≤50	51,200	256 / ≤50	10.3
GPU seconds (estimate)	Primary	16.8	2,048 / ≤50	14.5	256 / ≤50	151.4
	Replication	17.1	2,048 / ≤50	13.9	256 / ≤50	105.9

4 Limits of these conclusions

The experiment uses one donor, one recipe and one model, with each target analysed separately. The bootstrap quantifies variation across readout dates for those targets. Variation across fine-tunes, donors and recipes remains unmeasured. Second seeds were run at the full budget only, leaving short-run advantages unassessed against seed variation.

The scalar-only shortfall applies to the tested search and norm cap (optimization details). The mechanism and the contribution of the donor repair's input vectors remain unresolved (control details). The cost projection uses the first fresh seed as its reference and depends on the tested grid (cost accounting).

The repair addresses collateral Dutch CE damage after a fine-tune that deliberately worsens German prediction. Safety alignment and transfer of a new skill remain outside the study's scope.

5 Conclusion

For this model and recipe, at the full fitting budget, training a donor-initialized repair recovered 11.3 pp [+10.0, +12.7] and 20.4 pp [+19.0, +21.7] more of the Dutch damage than the tested scalar-only search over the donor repair's fixed directions, including its converged full-batch candidates. The gap was nearly as large at 256 and 512 tokens. Donor initialization helped most with little fitting data and few steps. At the full budget, its advantage over fresh fitting was positive but inside the ±2 pp equivalence band, and comparable to the difference between two fresh seeds. On the tested grid, its cheapest reaching cell used one eighth as many fit-token presentations as fresh fitting's cheapest reaching cell. When cost was minimized in unique tokens instead, both families reached the reference with 256 fit tokens.

AI assistance. I used OpenAI Codex and Anthropic Claude for research planning, code development, data analysis, and manuscript preparation. Scientific Agent Skills (Kassis et al. 2026) supplied the scientific-writing, scientific-critical-thinking, peer-review and citation-management workflows used to review organization, claims and references.

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A Method details

Model and fine-tunes. The base model has 66,931,200 parameters (residual width 768, MLP width 3,072). The 24 matrices are the query, key, value, output and two MLP matrices of each layer. Each fine-tune trains with Adam (Kingma and Ba 2015) at learning rate 3e-6 for 200 steps on a hinge term with a 1-nat margin, $\max(0, 1 - \Delta\mathrm{CE}_{\mathrm{de}})$ per window, and a KL term of weight 1 on the retain languages. Its checkpoint is chosen on German validation dates.

Repairs. Families initialized from the donor repair use $\min(1, \|D\|_F / \|E_{\mathrm{donor}}\|_F)\, E_{\mathrm{donor}}$. Fresh input factors have variance $1/d_{\mathrm{in}}$. Every family uses 10 warm-up steps, followed by a constant learning rate, with gradient clipping at 1.0. Saving a checkpoint every 50 steps at four learning rates (Table 5) gives 48 candidates.

Table 5: Learning-rate grids of the repair families.

Family	Learning rates
Scalar coefficients only	3e-3, 1e-2, 3e-2, 1e-1
Donor initialization, donor initialization output only, random initialization and fresh	1e-4, 3e-4, 1e-3, 3e-3

The learning-rate grids stayed fixed for the 50-step runs. Several output-only cells and the donor-initialized replication cell at 512 tokens select the first checkpoint at the smallest learning rate. Repeated values across step caps can therefore arise from checkpoint reuse alone. Advantages over fresh and random initialization change nonmonotonically as the step cap or budget increases.

Random initialization compares initial directions at matched norm, with both factors subsequently trained. The current controls do not compare the donor's trained input vectors with frozen random input vectors while training only the output factor.

Selection. A candidate is feasible if the ratio of German damage with and without the repair lies between 0.9 and 1.1, its German KL drift from the fine-tuned model is at most 0.05 times the base model's German KL from the fine-tune, and it moves Dutch toward the base model. Feasible candidates are ranked by Dutch recovery, then by smaller norm. Of the top five, the first that raises no retain language's CE by more than 0.01 nat/token is chosen.

Data. Portuguese, a seventh language, is measured as a diagnostic on the selection and readout dates. Its selection windows are included in the selection-token and evaluation costs. Its role is diagnostic only; it is excluded from the loss and selection constraints.

The data check would have excluded a readout date if a scan found it in the project's data, pilot or run files, or if its text window duplicated text already in use. All dates passed. The scan covers recorded project use; prior model exposure to these texts remains unresolved. Selection uses 32 dates and readout uses 92 dates. Each date contributes one 257-token window (256 next-token predictions) from the same turn in every language. Both targets use the same 92 readout dates. The fit windows and selection dates are separate from the donor repair's. The fit budgets are nested. At the 256-token budget, one Dutch window contributes 128 predictions, one German window 64, and four retain windows 16 each. The 512-token budget doubles these counts. At 2,048 tokens, four Dutch and two German windows contribute 256 predictions each, and one retain window contributes 128 per language.

Study record. According to my records, the rules, margins and families were fixed in a dated protocol before the readout data were built and the target fine-tunes were run. One author-dated amendment, made before any target fine-tune or readout, fixed the readout data and clarified the wording of the primary interpretation while keeping the decision rules unchanged. The protocol, amendment and project log remain unpublished and lack independent timestamps, so their chronology rests on the author's records rather than verifiable preregistration.

Uncertainty. The joint bootstrap uses 5,000 date resamples shared across both blocks of a target. Each resample recomputes the mean damages and then takes their recovery ratios. The ratio is therefore formed at the aggregate level. The readout constraints are applied to the point estimates on all dates. Equivalence is assessed for the five contrasts marked in Table 3; equivalence between donor and random initialization remains unestablished at every budget and step cap.

B Checks behind the primary reading

Table 6 lists recovery for every 2,048-token cell and the fixed donor repair, together with the quantities constrained during selection, now evaluated on the readout dates. All satisfy the constraints. On its own fit tokens, the chosen scalar-only repair removes 87.0% and 79.2% of the fine-tune's Dutch KL from the base model. Every family that trains the factors removes at least 98.4%.

Table 6: Readout of the 2,048-token cells (up to 600 steps) and the fixed donor repair on the 92 readout dates. Rows marked '+ full-batch optimization' also include the two converged candidates. KL recovery (%) is the share of the fine-tune's Dutch KL from the base model that the repair removes. German effect kept is the ratio of German CE damage with and without the repair (required 0.9–1.1). German drift is the repair's German KL from the fine-tuned model relative to the base model's KL from the fine-tuned model (required at most 0.05). Worst retain change is the largest CE change in English, French, Spanish or Italian against the fine-tuned model in nat/token (required at most +0.01).

Target	Repair	Recovery (%) [95% CI]	KL recovery	German effect kept	German drift	Worst retain change
Primary	Scalar coefficients only	86.9 [85.5, 88.3]	86.7	1.005	0.0003	+0.0003

Target	Repair	Recovery (%) [95% CI]	KL recovery	German effect kept	German drift	Worst retain change
Primary	Scalar coefficients only + full-batch optimization	86.9 [85.5, 88.3]	86.7	1.005	0.0003	+0.0003
Primary	Donor initialization	98.2 [97.6, 98.9]	98.3	0.996	0.0005	−0.0007
Primary	Donor initialization, output only	97.1 [96.4, 97.7]	98.1	1.003	0.0003	−0.0006
Primary	Random initialization	97.6 [97.0, 98.3]	98.2	0.988	0.0005	−0.0009
Primary	Random initialization, second seed	97.2 [96.5, 98.0]	97.5	0.990	0.0005	−0.0012
Primary	Fresh	97.2 [96.5, 97.8]	97.7	0.990	0.0005	−0.0014
Primary	Fresh, second seed	98.2 [97.5, 98.9]	97.8	0.985	0.0006	−0.0015
Primary	Fixed donor repair	86.3 [84.8, 87.9]	84.4	0.999	0.0002	−0.0001
Replication	Scalar coefficients only	77.3 [75.7, 79.0]	76.5	1.007	0.0002	+0.0003
Replication	Scalar coefficients only + full-batch optimization	78.4 [76.7, 80.0]	77.8	1.007	0.0003	+0.0004
Replication	Donor initialization	98.8 [98.1, 99.4]	98.6	1.001	0.0003	−0.0005
Replication	Donor initialization, output only	97.4 [96.8, 98.1]	98.2	1.007	0.0002	−0.0006
Replication	Random initialization	98.7 [98.0, 99.4]	97.9	0.990	0.0004	−0.0010
Replication	Random initialization, second seed	97.5 [96.8, 98.3]	97.8	0.996	0.0004	−0.0009
Replication	Fresh	97.4 [96.7, 98.0]	98.1	0.994	0.0004	−0.0012
Replication	Fresh, second seed	98.0 [97.4, 98.6]	98.4	0.990	0.0004	−0.0012
Replication	Fixed donor repair	58.6 [55.6, 61.4]	48.0	1.009	0.0003	−0.0001

The additional full-batch optimization uses coordinates $e_m = c_m\|b_m\|\|a_m\|$, in which $\|E\|_F = \|e\|_2$. The norm cap is therefore a ball, and radial scaling is its exact projection. Optimization uses the spectral step of Barzilai and Borwein (1988) in the projected-gradient scheme of Birgin et al. (2000) with a monotone Armijo line search, for at most 300 iterations. It runs from the initial scalar coefficients and from the checkpoint with the lowest fit loss ("best checkpoint").

A run counts as converged only if all losses and gradients, including line-search trials, are finite and the training and export losses agree within 1e-5 nat/token at the start. It must also stop with a projected-gradient norm at most 1e-4 of its initial reference, a relative fit-loss decrease below 1e-7 for five iterations, or the line-search floor with stationarity at most 1e-3 of that reference. Both starts use the projected-gradient reference from the initial scalar coefficients. Agreement between their final losses is reported descriptively, separately from the convergence criteria.

Table 7: Full-batch scalar optimization converged from both starts on both targets. The end losses differ by a relative 2.0e-5 on the primary and 1.1e-6 on the replication target.

Target	Start	Stop	Iterations	Fit loss, start → end	Stationarity / reference
Primary	initial scalar coefficients	relative decrease	107	0.02516 → 0.02070	0.0062
Primary	best checkpoint	relative decrease	84	0.02129 → 0.02070	0.0019
Replication	initial scalar coefficients	line-search floor	69	0.08323 → 0.03682	0.0002
Replication	best checkpoint	relative decrease	49	0.03870 → 0.03682	0.0004

The scalar-only shortfall applies to the tested search: four learning rates for 600 steps and optimization to convergence from two starts under the norm cap $\|E\|_F \le \|D\|_F$. Other assignments of the 24 scalar coefficients and larger repair norms remain untested.

C Cost accounting

With C^* a family's cheapest reaching cost, found separately in each currency, reuse pays off for more than

$$K^* = \frac{C_{\text{donor}}}{C^*(\text{fresh}) - C^*(\text{donor initialization})}$$

targets (Table 4). In fit-token presentations, both cheapest cells use the shortest step cap in the grid, and donor initialization uses its smallest budget. The one-eighth ratio is therefore the budget ratio 2,048/256, set by the tested grid. With the donor repair's cost of 3,686,400 presentations, this gives $K^* \approx 10.3$. The calculation amortizes the measured per-target costs. The 11-target break-even is a cost projection; the experiment used two targets. The accounting exclusions are detailed below.

Every donor-initialized cell reaches the reference, down to 256 tokens and 50 steps. Fresh fitting reaches it with 2,048 tokens at every step cap, but with 256 or 512 tokens only at step caps of 200 to 600. No scalar-only cell reaches it, and neither does the fixed donor repair (Figure 3).

A random start saves no presentations on the primary target. On the replication target, it reaches the reference with fewer presentations than fresh fitting, $K^* = 18$, charged with the full donor cost although it uses only the donor repair's norm.

Unique tokens count a cell's fit tokens plus the 57,344 predicted tokens on the selection dates (32 dates × 7 languages × 256). Fit-token presentations count the optimizer steps of all four learning rates up to the cap multiplied by the unique fit tokens. GPU seconds count training seconds prorated to the cap, plus the cell's share of selection evaluation according to its number of candidates. For the scalar-only search with full-batch optimization, presentation and GPU-second costs include both additional optimization runs and the evaluation of 48 checkpoints used to choose the second start.

No cell includes the readouts of the base and fine-tuned models or the damage check before fitting. Presentations also exclude the forward passes used for selection. The donor repair's one-time cost is 131,072 unique tokens (its 73,728 fit tokens and the selection tokens), 3,686,400 presentations (3 learning rates × 200 steps × 6,144 tokens per minibatch), and an estimated 336 GPU seconds for training, selection and readout. The donor fine-tune is excluded because, in use, the donor model already exists.

The reference's paired interval against itself is [0, 0] by construction. At 256 fit tokens with a 600-step cap, fresh fitting reaches the reference narrowly: its interval's lower end is −1.985 pp on the primary target and −1.957 pp on the replication target, against the −2 pp threshold. This cell sets the lowest unique-token cost for fresh fitting.

The break-even calculation uses the first fresh seed as its reference, with its size set by the tested grid (Section 3.3). Sensitivity to a different reference seed remains unevaluated.

D Reproducibility and availability

Code, saved measurements and reproduction instructions are on GitHub. The guide covers replaying the analysis and running the available model procedures.